Email Spam Classification

Classify each email as normal (not spam) or spam using K-Nearest Neighbors and a linear Support Vector Machine.

Objective

Build a binary classifier for the Kaggle emails.csv dataset and compare KNN and SVM performance. Spam is the positive class (label 1); normal email is label 0.

Dataset at a glance

Item	Description
Source	Email Spam Classification Dataset, Kaggle (Balaka18).
Records	5,172 emails; no missing values or duplicate rows reported.
Input columns	3,000 pre-tokenized word-frequency features plus an email identifier.
Target	Prediction: 0 = not spam (3,672), 1 = spam (1,500).
Train/test	4,137 training rows and 1,035 test rows; stratified 80/20 split.

Dataset source: kaggle.com/datasets/balaka18/email-spam-classification-dataset-csv

The email identifier is not a predictive feature. TF-IDF is fitted inside each model pipeline so it learns from training rows only.

Pre-processing and model setup

Both classifiers use the same stratified split and the same training-only text weighting step.

Pre-processing

Duplicate records were removed if present, the Email No. identifier was dropped, and word-frequency columns were converted to numeric values. The documented binary target labels were retained. The class distribution is imbalanced, so the test split was stratified to preserve the class proportions.

Label	Meaning	Rows
0	Not spam	3,672
1	Spam	1,500
Total	All emails	5,172

Models

K-Nearest Neighbors: k=5, cosine distance, distance-weighted voting. Cosine distance compares the direction of sparse word vectors. **Linear SVM:** LinearSVC with C=1.0 and balanced class weights, which gives more weight to the minority spam class during training.

```
models = {
  "KNN (k=5)": make_pipeline(
      TfidfTransformer(),
      KNeighborsClassifier(n_neighbors=5, metric="cosine",
                           weights="distance", algorithm="brute")),
  "Linear SVM": make_pipeline(
      TfidfTransformer(),
      LinearSVC(C=1.0, class_weight="balanced",
                random_state=42, max_iter=10000))
}
```

Accuracy measures the fraction of correct predictions. Spam precision measures the reliability of spam flags; spam recall measures how many actual spam emails are detected. F1 balances spam precision and recall.

MODEL COMPARISON

Held-out classification results

Both models were evaluated on the same 1,035 test emails, including 300 spam messages.

Performance summary

Model	Accuracy	Balanced accuracy	Spam precision	Spam recall	Spam F1
KNN (k=5)	0.9140	0.9109	0.8187	0.9033	0.8590
Linear SVM	0.9729	0.9780	0.9224	0.9900	0.9550

Confusion matrices

Rows are actual labels and columns are predicted labels. The positive class is spam (1).

KNN (k=5)	Predicted not spam	Predicted spam
Actual not spam	675	60
Actual spam	29	271

Linear SVM	Predicted not spam	Predicted spam
Actual not spam	710	25
Actual spam	3	297

Linear SVM is the stronger model on this test set. It detected 297 of 300 spam emails and had the highest spam F1 (0.9550), while misclassifying 25 normal messages as spam.

Conclusion

The Linear SVM had higher accuracy, balanced accuracy, spam precision, spam recall, and spam F1 than KNN. It is the better choice for these results. The operating threshold should still reflect the cost of false positives (normal messages flagged) versus false negatives (missed spam).